

HTML5 – Day – 3 Hands on Activities – Vasundhara

Problem 1

Assessment Goal: Check ability to apply styling and understand selectors and box model.

Hands-on Tasks:

1. Create an external CSS file and link it to HTML
2. Apply different text styles to headings and paragraphs
3. Add background colour and borders to a section
4. Demonstrate margin and padding on a div
5. Style the form created on Day 2

Expected Outcome:

A visually styled webpage using external CSS and proper selectors.

Source Code:

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>Student Registration</title>
6      <link rel="stylesheet" href="style.css">
7  </head>
8  <body>
9
10 <header>
11     <h1>Student Registration</h1>
12 </header>
13 <center>
14 <p>
15     Lorem ipsum dolor sit, amet consectetur adipisicing elit.
16     Blanditiis sed nostrum soluta tempora quo et. Nam deserunt
17     molestiae obcaecati laudantium quo delectus consequuntur magnam
18     nulla dignissimos repellat? Explicabo, at omnis?
19 </p>
20 </center>
21 <section class="info-section">
22     <h2>About Registration</h2>
23     <p>
24         Students must enter correct details before submitting the form.
25     </p>
26 </section>
27
28 <section class="form-section">
```

Output:

This box shows margin and padding clearly.

Name:

Email:

Phone:

Age:

Date of Birth:


Register

Student Registration

Lorem ipsum dolor sit, amet consectetur adipisicing elit. Blanditiis sed nostrum soluta tempora quo et. Nam deserunt molestiae obcaecati laudantium quo delectus consequuntur magnam nulla dignissimos repellat? Explicabo, at omnis?

About Registration

Students must enter correct details before submitting the form.

This box shows margin and padding clearly.

Name:

Email:

Explanation:

This task uses HTML anchor tags and the `<nav>` element to enable navigation between multiple pages. Form inputs use HTML5 attributes like `required` and `placeholder` for basic client-side validation.

Problem 2: Personal Profile Card (Level-1)

Scenario

You are building a simple personal profile card for a portfolio website. The card will display a user's name, role, short bio, and contact button, styled purely using CSS.

Requirements

Create a profile card layout containing:

- Name (heading)
- Role/title
- Short description paragraph
- Contact button

Apply:

- Text formatting (alignment, decoration)
- Font styles (font-family, size, weight)
- Typography enhancements (line-height, letter-spacing)

Use the CSS Box Model:

- Padding for spacing inside the card
- Border to define the card boundary
- Margin to separate the card from the page edges

Add:

- Background color
- Border radius
- Hover effect on the button

Technical Constraints

- Use only HTML + CSS
- No images or icons
- No frameworks (Bootstrap, Tailwind, etc.)
- Inline CSS not allowed (use `<style>` or external CSS)

Learning Outcome

You should be able to:

- Understand how CSS styles text and fonts
- Apply box model concepts effectively
- Style elements using colors, borders, and spacing
- Improve visual hierarchy using typography

Source Code:

```
basic > day3h1.html > day3h2.html > <> index.html > html > body
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>Personal Profile Card</title>
6      <link rel="stylesheet" href="style.css">
7  </head>
8  <body>
9
10     <div class="profile-card">
11         <h1>Vasu</h1>
12         <h2>Web Developer</h2>
13         <p>
14             I am a beginner web developer focused on building
15             clean and responsive websites using HTML and CSS.
16         </p>
17         <button>Contact Me</button>
18     </div>
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20 </body>
21 </html>
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```

```

28 .profile-card h2 {
29     font-weight: normal;
30     color:  rgb(243, 83, 83);
31     margin-bottom: 10px;
32 }
33
34
35
36 .profile-card p {
37     font-size: 14px;
38     line-height: 1.6;
39     margin-bottom: 15px;
40 }
41
42
43 .profile-card button {
44     padding: 8px 15px;
45     background-color:  #7fefb3;
46     color:  white;
47     border: none;
48     border-radius: 5px;
49     cursor: pointer;
50 }
51
52
53 .profile-card button:hover {
54     background-color:  #e15d24;
55 }
56 header p {
57     border: 2px solid  #e41480;

```

```

49     cursor: pointer;
50 }
51
52
53 .profile-card button:hover {
54     background-color:  #e15d24;
55 }
56 header p {
57     border: 2px solid  #e41480;
58     padding: 10px;
59     margin: 15px auto;
60     width: 80%;
61     border-radius: 5px;
62     color:  #555;
63     text-align: center;
64 }
65

```


Output:



Explanation:

This problem demonstrates semantic navigation using the `<nav>` element with relative links.

It helps users move seamlessly between pages without using CSS frameworks or JavaScript.

Problem 3: Product Feature List Section (Level-1)

Scenario

You are designing a feature highlight section for a product landing page that lists key features in a visually appealing way.

Requirements

Create a section with:

- Section heading
- List of 4–5 features

Style the list using:

- Custom fonts
- Text color and background color
- Border and padding for each feature item

Use:

- CSS selectors (element, class, descendant)

Apply hover effects on feature items

Technical Constraints

- Use unordered list ()
- No JavaScript
- No layout frameworks
- Only basic HTML structure

Learning Outcome

You will be able to:

- Practice CSS selectors
- Style lists professionally
- Improve UI consistency

Source Code:

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>Product Features</title>
6
7      <style>
8          /* Element selector */
9          body {
10              font-family: Arial, sans-serif;
11              background-color: #f2f2f2;
12          }
13
14          /* Class selector */
15          .feature-section {
16              width: 60%;
17              margin: 40px auto;
18              background-color: #ffffff;
19              padding: 20px;
20          }
21
22          /* Descendant selector */
23          .feature-section h2 {
24              text-align: center;
25              color: #2c3e50;
26              font-family: "Trebuchet MS", sans-serif;
27          }
28
29          /* Descendant selector */
```

```
3  </head>
7      <style>
30          .feature-list li {
31              list-style-type: none;
32              background-color: #e8f4ff;
33              color: #003366;
34              border: 2px solid #3498db;
35              padding: 12px;
36              margin: 10px 0;
37              font-family: Verdana, sans-serif;
38              transition: 0.3s;
39          }
40
41          /* Hover effect */
42          .feature-list li:hover {
43              background-color: #3498db;
44              color: white;
45          }
46      </style>
47  </head>
48
49  <body>
50
51  <section class="feature-section">
52      <h2>Product Features</h2>
53
54      <ul class="feature-list">
55          <li>High Performance</li>
```

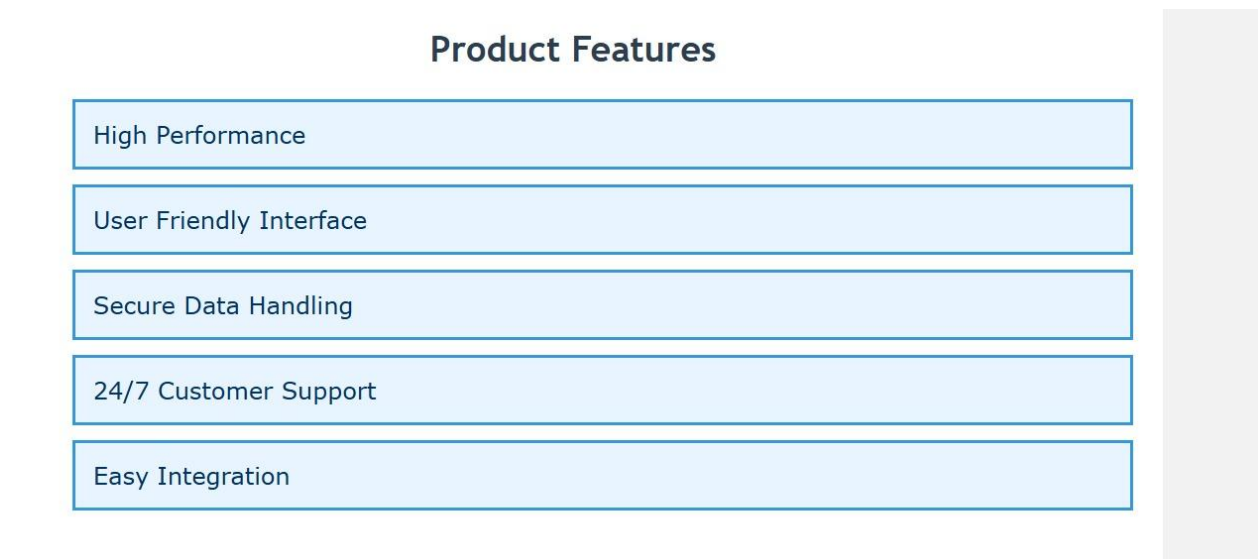


```

51  <section class="feature-section">
52      <h2>Product Features</h2>
53
54  <ul class="feature-list">
55      <li>High Performance</li>
56      <li>User Friendly Interface</li>
57      <li>Secure Data Handling</li>
58      <li>24/7 Customer Support</li>
59      <li>Easy Integration</li>
60  </ul>
61 </section>
62
63 </body>
64 </html>

```

Output:



Explanation:

This section uses an unordered list styled with CSS element, class, and descendant selectors. Hover effects and box model properties enhance visual appearance and UI consistency.

Problem 4: Responsive Blog Layout (Level-2)

Scenario

You are creating a blog homepage that displays articles in a modern responsive layout suitable for desktop, tablet, and mobile screens.

Requirements

Create a page layout with:

- Header
- Main content area with multiple blog cards

Use:

- **Flexbox or CSS Grid** for layout
- Box model for spacing

Each blog card should contain:

- Title
- Short description
- Read More button

Apply:

- Background colors
- Borders
- Typography enhancements

Use **media queries** to:

- Display multiple columns on desktop
- Switch to single column on mobile

Technical Constraints

- Only HTML + CSS
- No CSS frameworks
- No fixed widths (use percentages or flexible units)

Learning Outcome

Learners should be able to:

- Understand Flexbox/Grid layouts
- Apply responsive design principles
- Write effective media queries
- Build real-world page layouts

Source Code:

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>Responsive Blog</title>
6
7      <style>
8          /* Element selector */
9          body {
10              margin: 0;
11              font-family: Arial, sans-serif;
12              background-color: #f4f4f4;
13          }
14
15          /* Header */
16          header {
17              background-color: #2c3e50;
18              color: white;
19              padding: 20px;
20              text-align: center;
21          }
22
23          /* Main container */
24          .blog-container {
25              display: flex;
26              flex-wrap: wrap;
27              gap: 20px;
28              padding: 20px;
29          }
```

```
32      .blog-card {
33          background-color: white;
34          border: 2px solid #ddd;
35          padding: 15px;
36          box-sizing: border-box;
37          flex: 1 1 30%;
38      }
39
40      /* Descendant selectors */
41      .blog-card h3 {
42          color: #333;
43      }
44
45      .blog-card p {
46          color: #555;
47      }
48
49      .blog-card button {
50          padding: 8px 12px;
51          background-color: #3498db;
52          color: white;
53          border: none;
54          cursor: pointer;
55      }
56
57      .blog-card button:hover {
```

```

56
57     .blog-card button:hover {
58         background-color: #1d6fa5;
59     }
60
61     /* Responsive: Mobile */
62     @media (max-width: 600px) {
63         .blog-container {
64             flex-direction: column;
65         }
66     }
67     </style>
68 </head>
69
70 <body>
71
72 <header>
73     <h1>My Blog</h1>
74     <p>Latest Articles</p>
75 </header>
76
77 <main class="blog-container">
78
79     <div class="blog-card">
80         <h3>Blog Post One</h3>

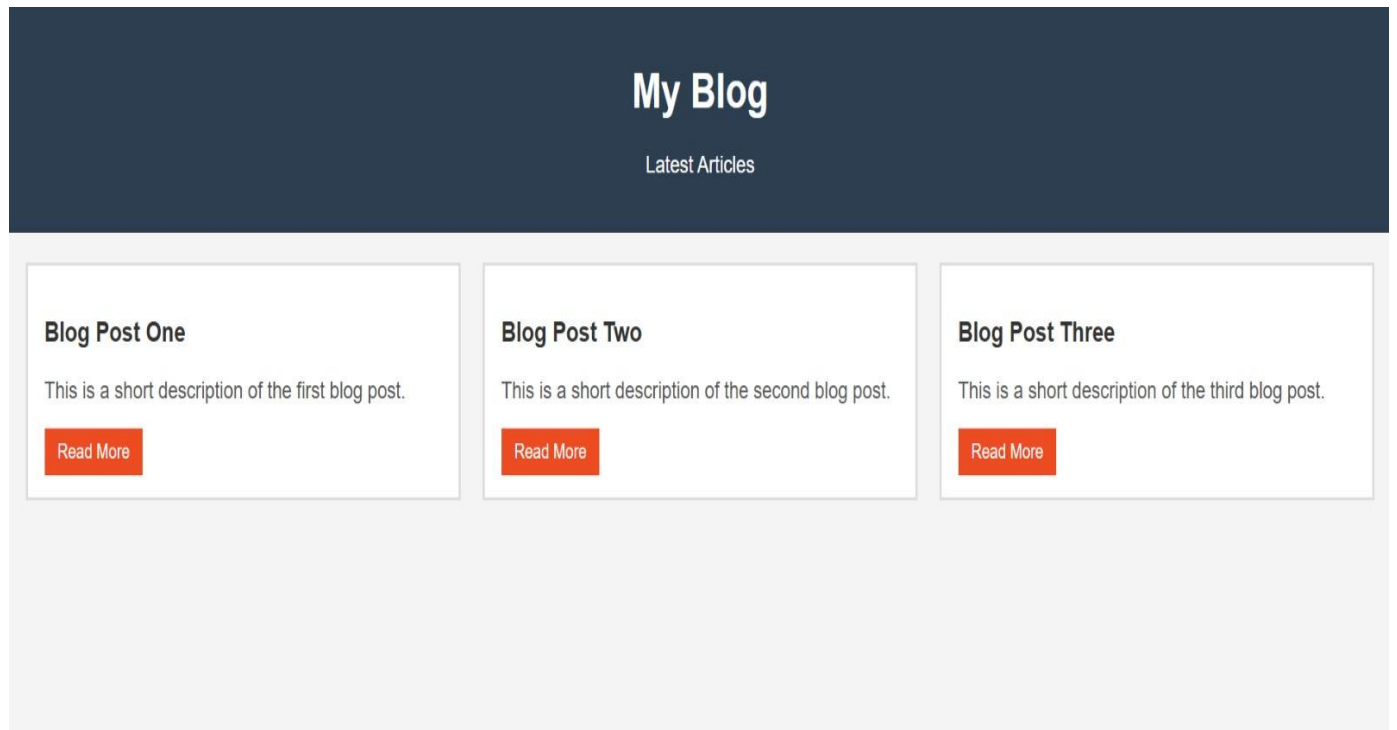
```

```

2 <html lang="en">
70 <body>
76
77 <main class="blog-container">
78
79     <div class="blog-card">
80         <h3>Blog Post One</h3>
81         <p>This is a short description of the first blog post.</p>
82         <button>Read More</button>
83     </div>
84
85     <div class="blog-card">
86         <h3>Blog Post Two</h3>
87         <p>This is a short description of the second blog post.</p>
88         <button>Read More</button>
89     </div>
90
91     <div class="blog-card">
92         <h3>Blog Post Three</h3>
93         <p>This is a short description of the third blog post.</p>
94         <button>Read More</button>
95     </div>
96
97 </main>
98
99 </body>
100 </html>
101

```

Output:



Explanation:

This layout uses CSS Flexbox to arrange blog cards responsively across different screen sizes.

Media queries adjust the layout to display multiple columns on desktop and a single column on mobile.

Problem 5: Responsive Dashboard Layout (Level-2)

Scenario

You are designing a simple admin dashboard layout for a web application.

Requirements

Create a layout with:

- Sidebar navigation
- Main content area
- Info cards inside the main section

Use:

- display properties
- Flexbox or Grid for page structure

Style:

- Sidebar with background color
- Cards with borders, padding, and margins

Apply **media queries** to:

- Convert sidebar into top navigation on mobile
- Stack cards vertically on small screens

Technical Constraints

- HTML + CSS only
- No JavaScript
- No external UI libraries

Learning Outcome

You will be able to:

- Master page layouts using Flexbox/Grid
- Apply modern responsive web design techniques
- Understand how layout adapts across devices
- Build industry-relevant UI structures

Source Code:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4     <meta charset="UTF-8">
5     <title>Responsive Dashboard</title>
6
7     <style>
8         /* Element selector */
9         body {
10             margin: 0;
11             font-family: Arial, sans-serif;
12         }
13
14         /* Page Layout */
15         .dashboard {
16             display: flex;
17             min-height: 100vh;
18         }
19
20         /* Sidebar */
21         .sidebar {
22             background-color: #2c3e50;
23             color: white;
24             padding: 20px;
25             flex: 1;
26         }
27
28         .sidebar h2 {
29             margin-top: 0;
30
31
32             .sidebar ul {
33                 list-style: none;
34                 padding: 0;
35             }
36
37             .sidebar li {
38                 margin: 15px 0;
39             }
40
41         /* Main Content */
42         .main-content {
43             flex: 4;
44             padding: 20px;
45             background-color: #f4f4f4;
46         }
47
48         /* Cards Container */
49         .cards {
50             display: flex;
51             flex-wrap: wrap;
52             gap: 20px;
53         }
54
```

```

56     .card {
57         background-color: white;
58         border: 2px solid #ddd;
59         padding: 20px;
60         flex: 1 1 30%;
61         box-sizing: border-box;
62     }
63
64     /* Responsive Design */
65     @media (max-width: 768px) {
66
67         /* Sidebar becomes top navigat
68         .dashboard {
69             flex-direction: column;
70         }
71
72         .sidebar {
73             text-align: center;
74         }
75
76         /* Cards stack vertically */
77         .cards {
78             flex-direction: column;
79         }
80     }
81 </style>

```

```

88 <aside class="sidebar">
89     <h2>Admin Panel</h2>
90     <ul>
91         <li>Dashboard</li>
92         <li>Users</li>
93         <li>Reports</li>
94         <li>Settings</li>
95     </ul>
96 </aside>
97
98 <main class="main-content">
99     <h2>Dashboard Overview</h2>
100
101     <div class="cards">
102         <div class="card">
103             <h3>Total Users</h3>
104             <p>1,200</p>
105         </div>
106
107         <div class="card">
108             <h3>Sales</h3>
109             <p>$8,500</p>
110         </div>
111
112         <div class="card">
113             <h3>Performance</h3>

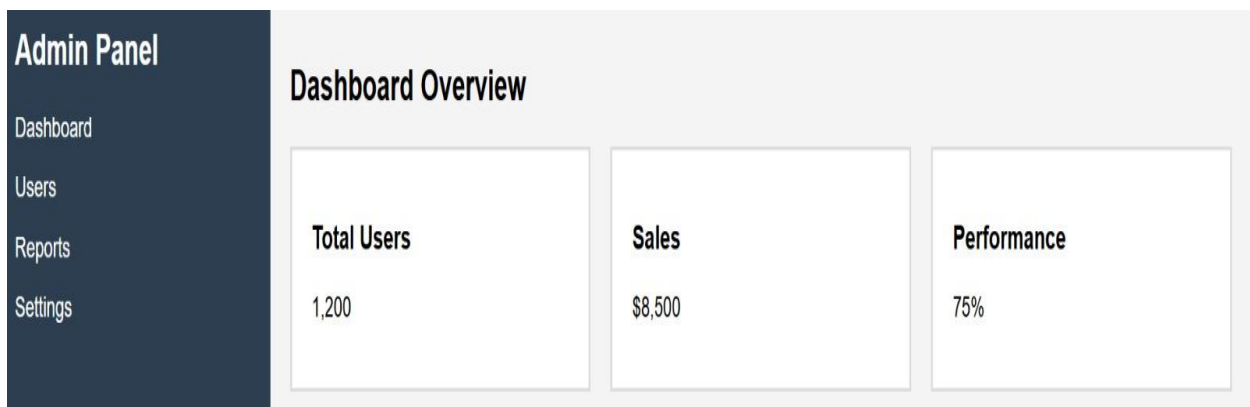
```

```

108         <h3>Sales</h3>
109         <p>$8,500</p>
110     </div>
111
112     <div class="card">
113         <h3>Performance</h3>
114         <p>75%</p>
115     </div>
116 </div>
117
118 </main>
119
120 </div>
121
122 </body>
123 </html>
124

```

Output:



Explanation:

This dashboard uses CSS Flexbox to create a sidebar and main content structure.

Media queries adapt the layout for mobile by converting the sidebar to top navigation and stacking cards vertically.